MEGA SUPERMART

Employee Absenteeism

Exploratory Data Analysis | 2025 HR Dataset

Executive Summary

This report examines employee absenteeism across departments, employment types and demographic groups using the 2025 Mega Supermart HR dataset.
The analysis applies descriptive statistics, data validation, Pearson and Spearman correlation, IQR outlier detection, Welch's independent-samples t-test and Chi-square testing to identify workforce attendance patterns and potential areas for HR investigation.

Dataset measure	Result
Total employees	1,200
Valid absence records	1,198
Dataset columns	17
Weighted absence rate	2.52%
Mean absence rate	2.71%
Median absence rate	1.73%
Mean absence hours	46.73
Median absence hours	32

Key findings

- Customer Service recorded the highest departmental weighted absence rate (3.38%), followed by Sales (3.08%).
- Part-time employees had a higher average absence rate (3.94%) than full-time employees (2.23%).
- Welch's t-test confirmed a statistically significant difference between employment groups ($p < 0.001$).
- A total of 63 employees (5.26%) exceeded the absence-rate IQR upper threshold of 7.91%.

Customer Service part-time employees had the highest departmental outlier rate (25.00%).

- Sales accounted for the largest number of high-absence cases (38 employees).
- Age and length of service showed statistically significant but weak negative associations with absence rates.

Overall, absenteeism is concentrated in particular workforce segments, particularly Customer Service and Sales.

1. Data Quality and Validation

1.1 Missing Values Assessment

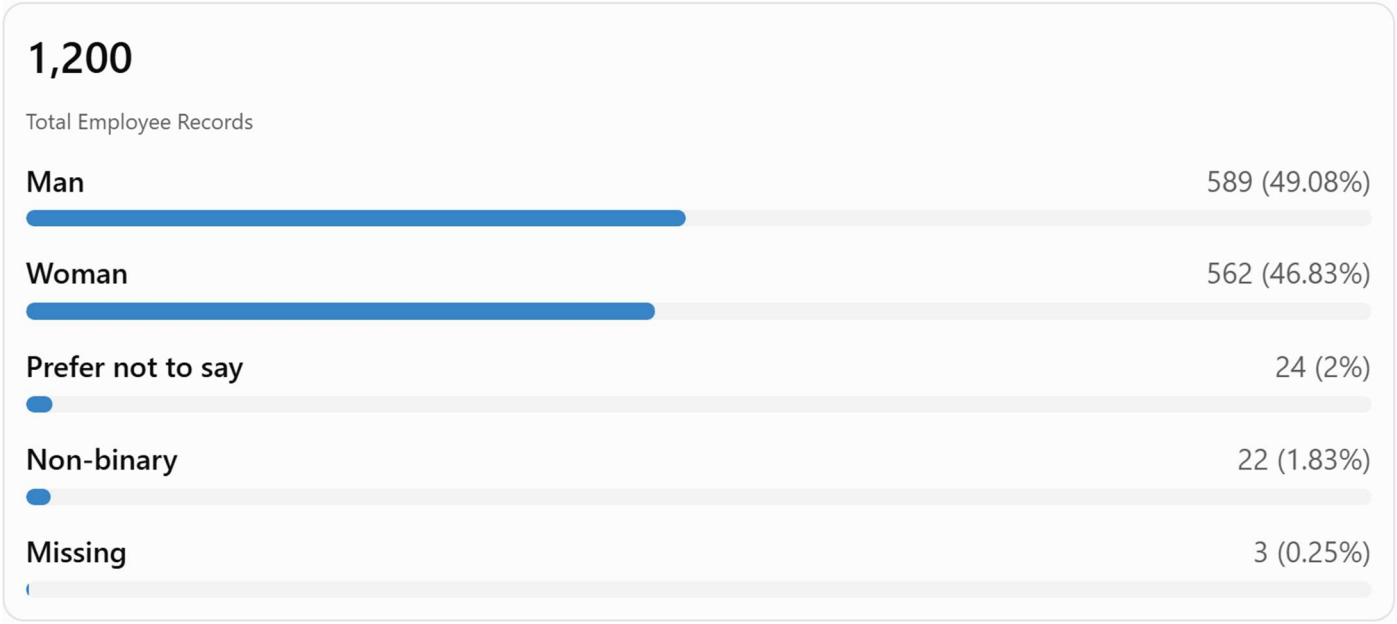
The dataset contains 1,200 employee records and 17 columns. Missing values were identified in three variables.

Figure 1.1.1

	Missing Count	Missing Percentage
Gender	3	0.25
AbsentHours	2	0.17
AbsenceRate	2	0.17

Five employee records are affected. The two records with missing attendance information have missing values in both AbsentHours and AbsenceRate. No imputation or deletion was performed. The 1,198 valid attendance records were used for subsequent absence analysis.
Inconsistent department naming conventions were also identified and require standardisation.

Figure 1.1.2



1.2 Absence Rate Validation

Absence rates were validated using the following formula:

AbsenceRate = AbsentHours / ScheduledHours2025

Figure 1.2.1

Validation measure	Result
Records tested	1,198
Matching records	1,198
Mismatching records	0
Validation accuracy	100%

All valid absence-rate records matched the underlying attendance calculations.

Figure 1.2.2

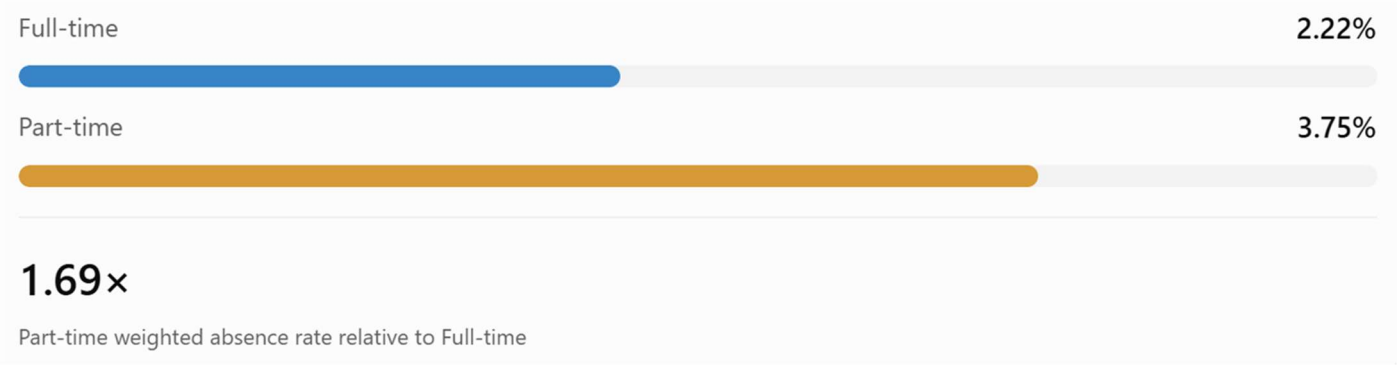


Figure 1.2.3

```
Q1: 12.0
Q3: 64.0
IQR: 52.0
Lower Bound: -66.0
Upper Bound: 142.0
Number of Outliers: 50
```

2. Overall and Departmental Absenteeism Analysis

2.1 Overall Absenteeism

The organisation's weighted absence rate was 2.52%, representing the proportion of scheduled working hours lost to absence.

The mean absence rate (2.71%) exceeded the median (1.73%). Similarly, mean absence hours (46.73) exceeded median absence hours (32). These differences indicate positively skewed distributions, suggesting that a relatively small group of employees with high absence levels influences overall averages.

2.2 Departmental Comparison

Department	Weighted absence rate	Average absence hours
Customer Service	3.38%	62.8
Sales	3.08%	57.5
Human Resources	2.49%	46.7
Operations	1.43%	26.2

Customer Service recorded approximately 2.36 times the weighted absence rate and 2.4 times the average absence hours of Operations.

The mean exceeded the median across all six departments. Customer Service and Sales showed the largest mean–median gaps, at 18.8 and 17.5 hours respectively.

Figure 2.2.1 Weighted Absence Rate by Department

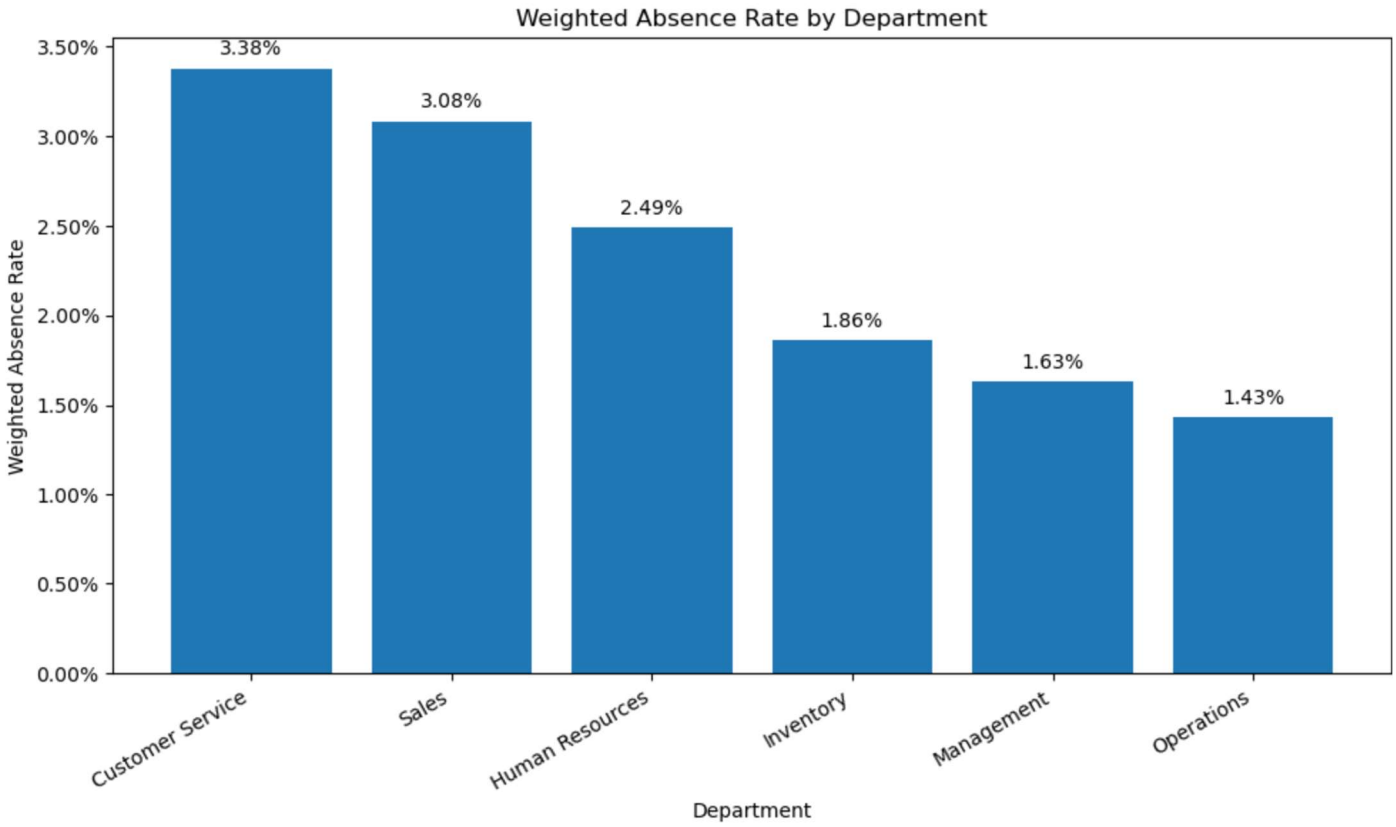


Figure 2.2.2 Average Absence Hours by Department

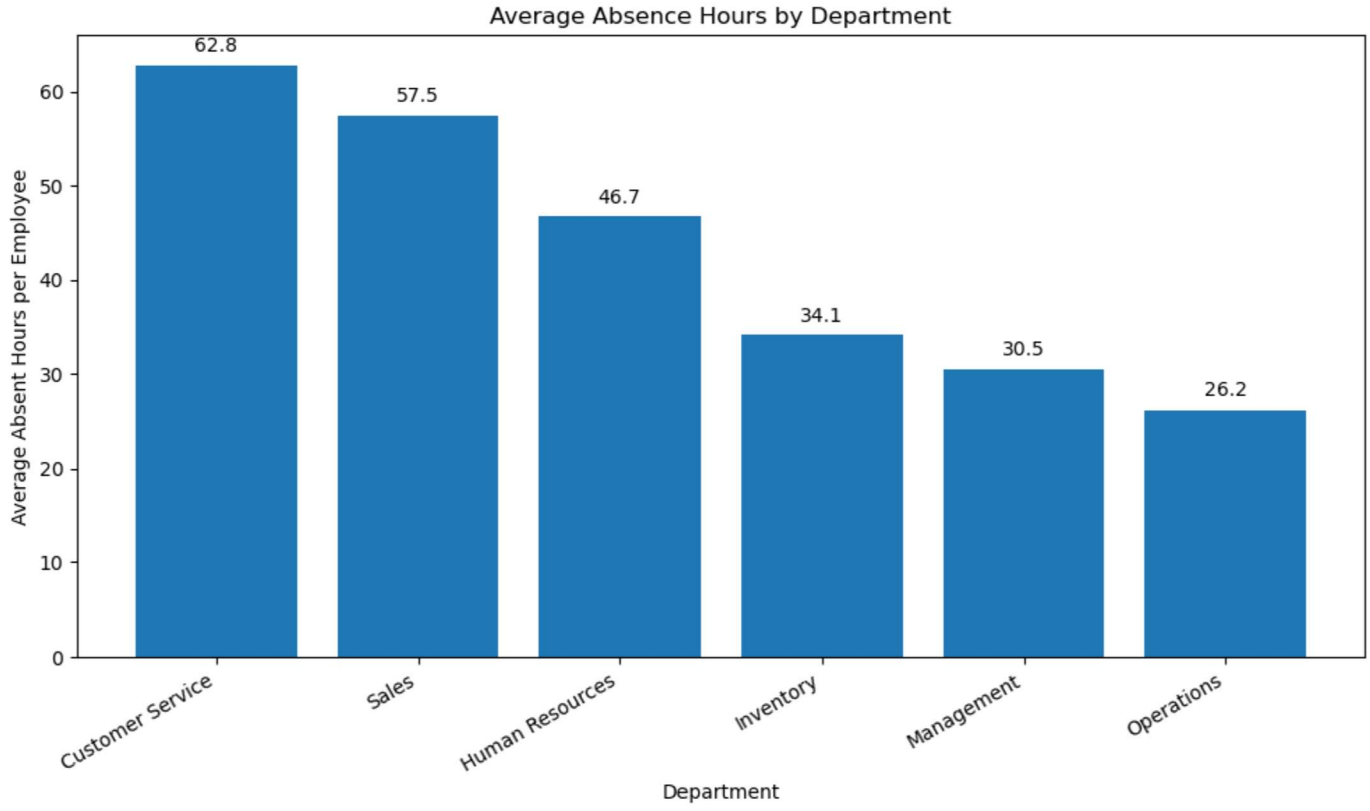


Figure 2.2.3 Mean vs. Median Absence Hours

Department	Mean Absent Hours	Median Absent Hours	Difference (Hours)
Customer Service	62.8	44.0	18.8
Sales	57.5	40.0	17.5
Human Resources	46.7	32.0	14.7
Inventory	34.1	28.0	6.1
Management	30.5	24.0	6.5
Operations	26.2	20.0	6.2

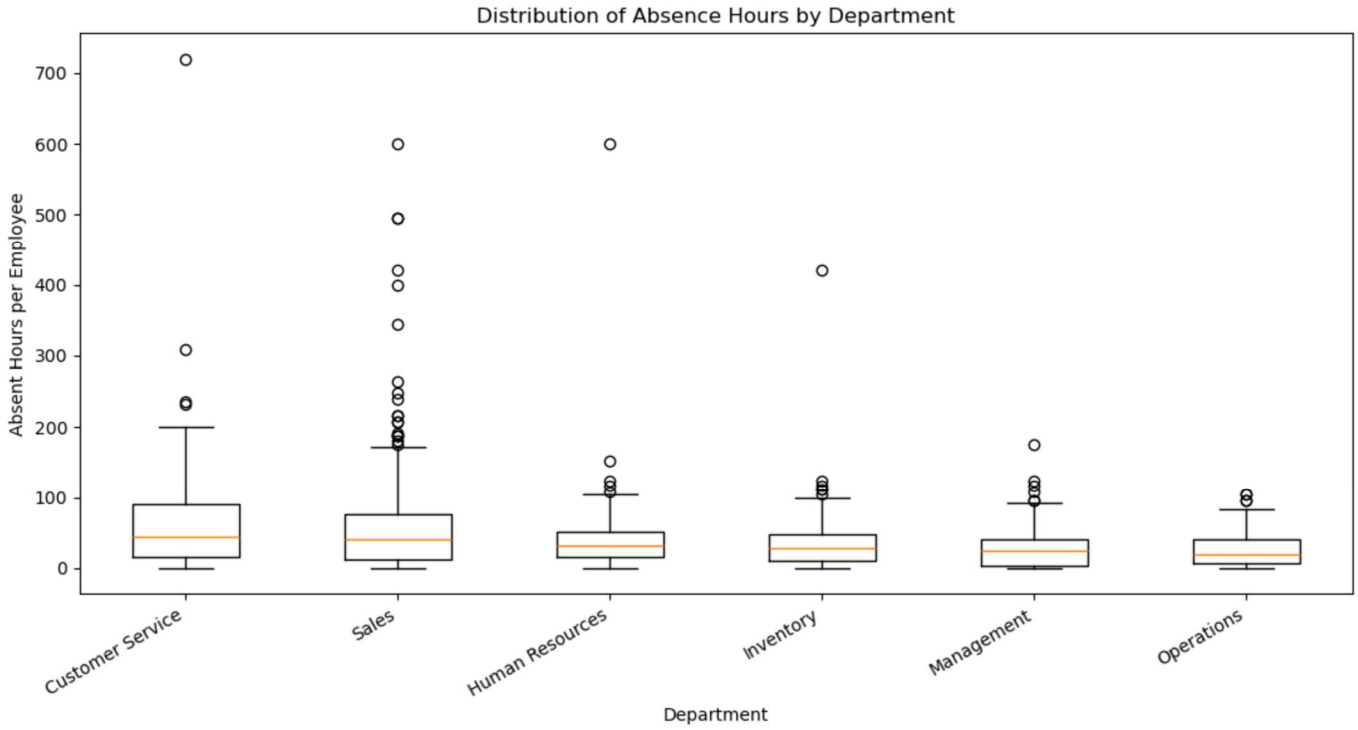
2.3 Departmental Absence-Hour Outliers

The earlier absence-hour IQR analysis identified 50 statistical outliers.

- Human Resources had the highest departmental outlier rate (7.4%).
- Sales recorded the largest number of outliers (19 employees).
- Management had the second-highest outlier rate (4.7%).
- Customer Service had the lowest outlier rate (2.1%).

These results identify potential areas for further investigation rather than necessarily indicating data errors or employee misconduct.

Figure 2.3 Department-Level Absence Outlier Analysis



3. Employee and Demographic Analysis

3.1 Top 10 Employees by Absence Hours

The employee-level analysis identified substantial differences in absence patterns.

- ACE01001 recorded the highest absence hours (720).

- Sales accounted for five of the top ten employees (50%).
- ACE00394 recorded the highest absence rate (57.69%).
- ACE00327 followed with an absence rate of 48.08%.
- Customer Service and Human Resources also had employees with notably high absence levels.

Employee reference	Absence hours	Scheduled hours	Absence rate
Employee 393	600	—	57.69%
Employee 1000 / ACE01001	720	2,080	34.62%
HR employee 326 / ACE00327	600	—	48.08%

All ten employees exceeded the overall absence-hour IQR upper boundary of 142 hours.

Figure 3.1.1 Top 10 Employees by Absence Hours

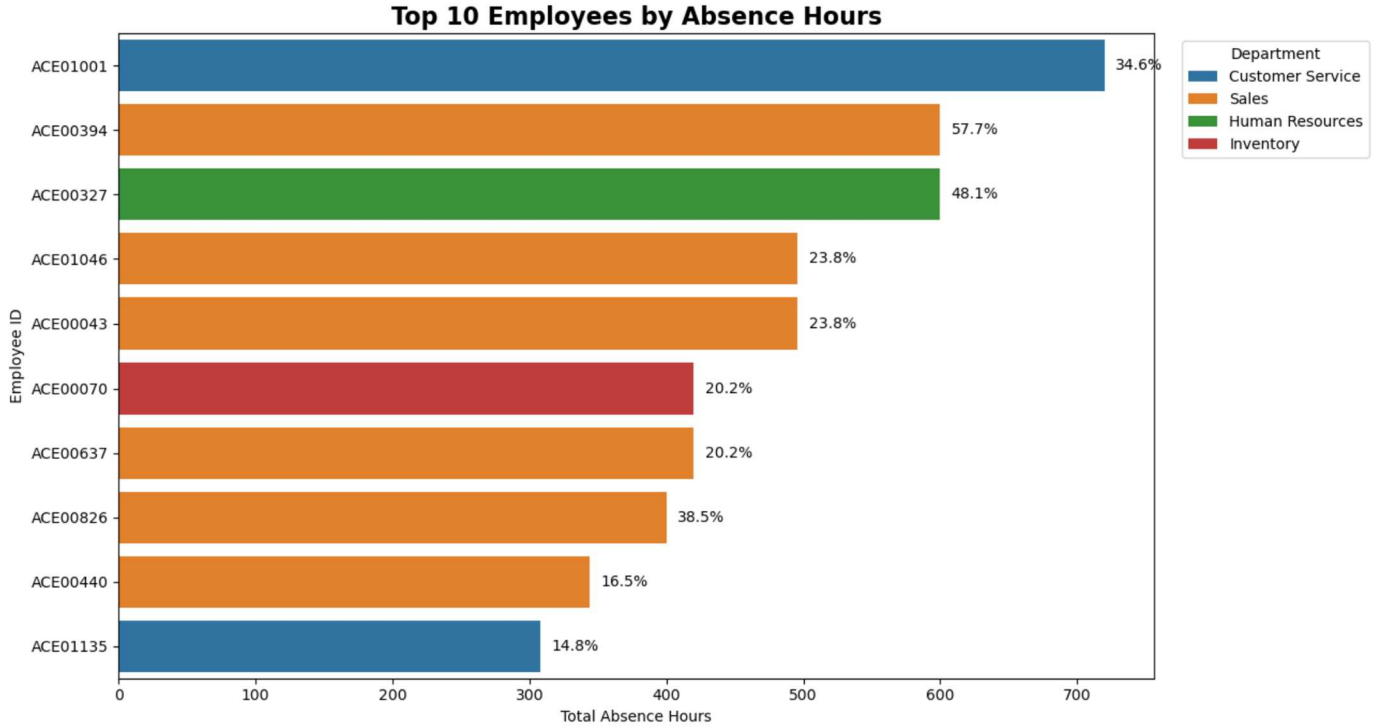


Figure 3.1.2

	EmployeeID	Department	JobTitle	ScheduledHours2025	AbsentHours	AbsenceRate	Outlier
1000	ACE01001	Customer Service	Customer Service Associate	2080	720.0	0.3462	True
393	ACE00394	Sales	Sales Supervisor	1040	600.0	0.5769	True
326	ACE00327	Human Resources	HR Business Partner	1248	600.0	0.4808	True
1045	ACE01046	Sales	Senior Sales Associate	2080	496.0	0.2385	True
42	ACE00043	Sales	Sales Associate	2080	496.0	0.2385	True
69	ACE00070	Inventory	Inventory Supervisor	2080	420.0	0.2019	True
636	ACE00637	Sales	Sales Associate	2080	420.0	0.2019	True
825	ACE00826	Sales	Senior Sales Associate	1040	400.0	0.3846	True
439	ACE00440	Sales	Sales Associate	2080	344.0	0.1654	True
1134	ACE01135	Customer Service	Customer Service Supervisor	2080	308.0	0.1481	True

3.2 Age Group Analysis

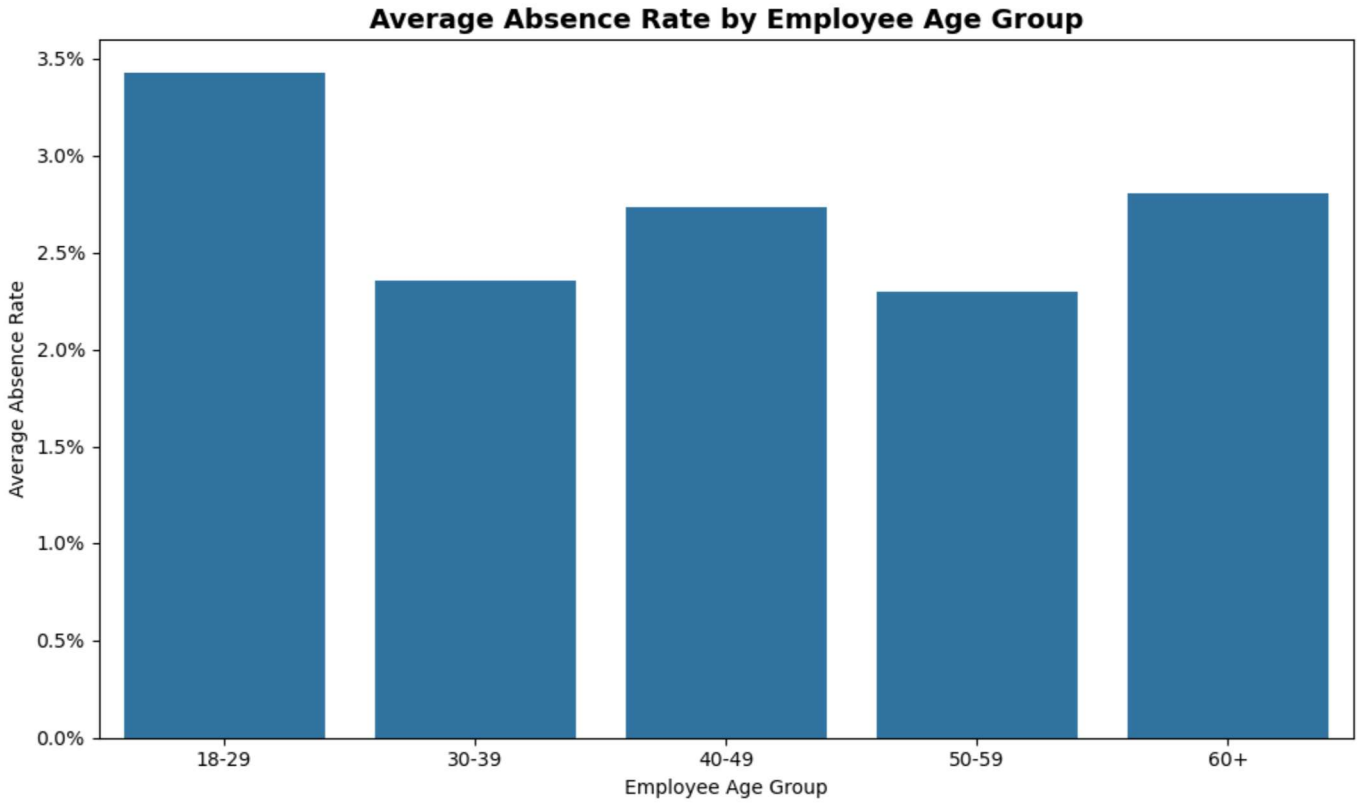
Age group	Average absence rate	Average absence hours
18–29	3.40%	57.6
30–39	2.40%	—
40–49	2.70%	—
50–59	2.30%	39.02
60+	2.80%	—

Employees aged 18–29 recorded the highest average absence rate and absence hours, while employees aged 50–59 recorded the lowest average absence rate. The 60+ group contains only 40 employees, so its results require cautious interpretation. The pattern is not linear, suggesting that age alone does not explain absenteeism.

Figure 3.2.1

```
ABSENCE ANALYSIS BY AGE GROUP
------------------------------------------------------------
AgeGroup  Employee_Count  Average_Absence_Rate  Median_Absence_Rate  Average_Absent_Hours  Median_Absent_Hours
   18-29            265                 0.034                0.020                57.600                 40.0
   30-39            378                 0.024                0.017                42.328                 32.0
   40-49            336                 0.027                0.017                46.857                 32.0
   50-59            179                 0.023                0.015                39.017                 32.0
     60+             40                 0.028                0.016                49.800                 30.0
```

Figure 3.2.2 Average Absence Rate by Employee Age Group



3.3 Gender Analysis

Gender	Average absence rate	Average absence hours
Male	2.80%	48.2
Female	2.70%	46.14
Non-binary	—	—
Prefer not to say	—	—

Median absence rates ranged from 1.5% to 1.9% across the four groups. The difference between male and female average absence rates was only 0.1 percentage points. The non-binary and prefer-not-to-say groups contained 22 and 24 employees respectively.
Overall, gender differences were relatively modest and should not be treated as evidence of an important independent absenteeism factor.

3.4 Length of Service Analysis

Length of service	Average absence rate	Average absence hours
Under 2 years	3.20%	56.33
2–5 years	—	—
5–10 years	—	—
10+ years	2.20%	35.71

Employees with less than two years of service recorded the highest absence rate (3.2%), while employees with ten or more years of service recorded 2.2%. Average absence hours decreased from 56.33 to 35.71, a reduction of approximately 36.6%.
Median absence rates also declined from 2.0% among newer employees to 1.5% among employees with five or more years of service.
These findings suggest a possible association between tenure and absenteeism. HR should consider onboarding, employee engagement and early-stage workforce support.

Figure 3.4.1 Average Absence Rae by Length of Service

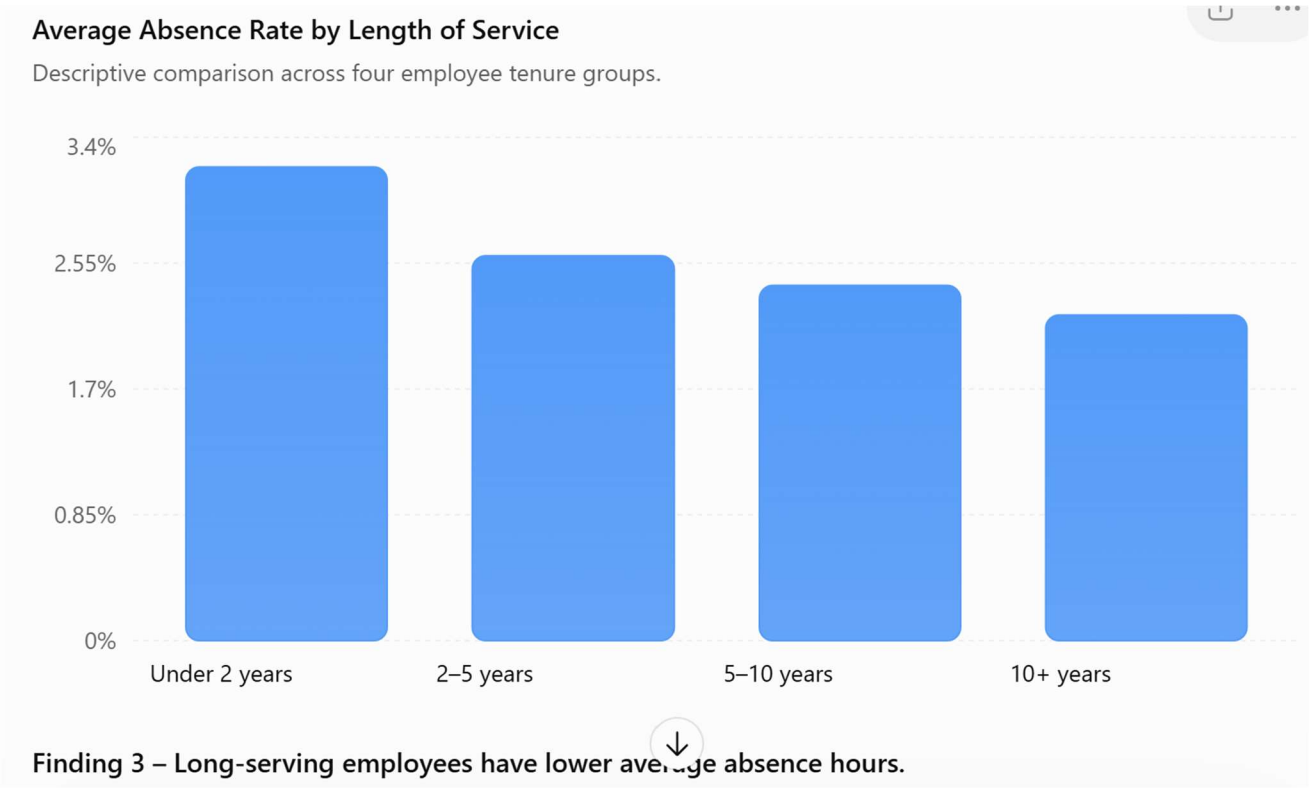


Figure 3.4.2 Absence Analysis by Length of Service

```
ABSENCE ANALYSIS BY LENGTH OF SERVICE
-------------------------------------------------------------
     ServiceGroup  Employee_Count  Average_Absence_Rate  Median_Absence_Rate  Average_Absent_Hours  Median_Absent_Hours
Less than 2 years             431                 0.032                0.020                56.325                 40.0
    2 to <5 years             359                 0.026                0.017                44.022                 32.0
   5 to <10 years             270                 0.024                0.015                40.652                 28.0
        10+ years             138                 0.022                0.015                35.710                 30.0
```

4. Employment Type Analysis

4.1 Absence Rate and Absence Hours

Measure	Full-time	Part-time
Mean absence rate	2.23%	3.94%
Median absence rate	1.50%	2.30%
Mean absence hours	46.3	47.85
Median absence hours	32	32

Part-time employees recorded a higher average absence rate by approximately 1.71 percentage points, despite having relatively similar average absence hours. This demonstrates the importance of using absence rates rather than absolute absence hours when comparing different employment arrangements.

Figure 4.1.1: Absenteeism Comparison by Employment Type

Metric	Full-time	Part-time
Number of employees	863	337
Average absenteeism hours	46.297	47.845
Median absenteeism hours	32.0	32.0
Average absenteeism rate	2.2%	3.9%
Median absenteeism rate	1.5%	2.3%

Figure 4.1.2: Average Absence Rate by Employment Type

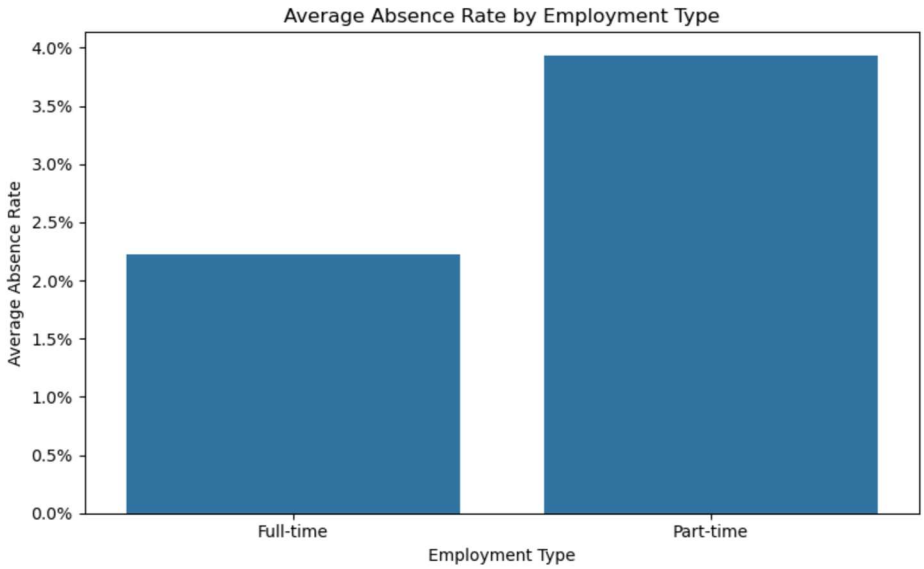
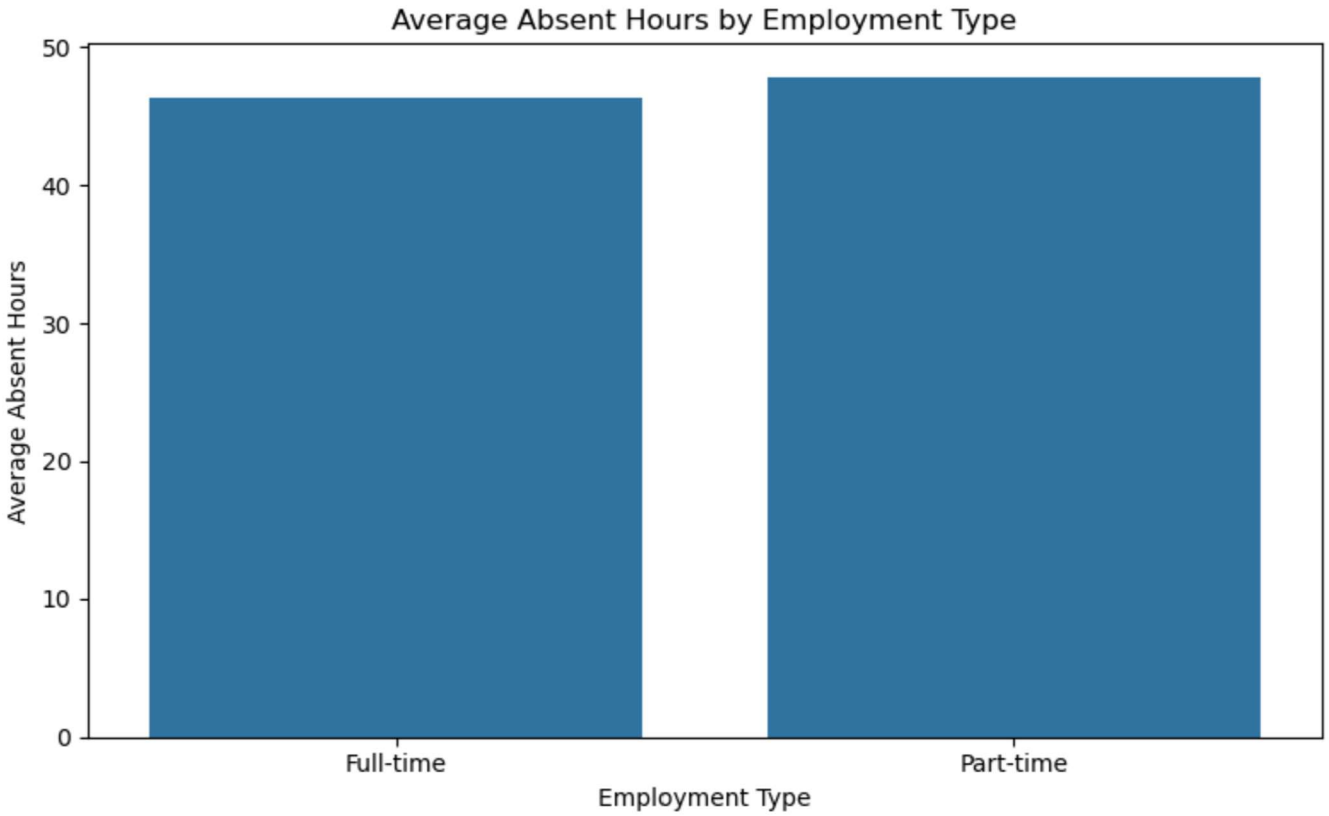


Figure 4.1.3: Average Absent Hours by Employment Type

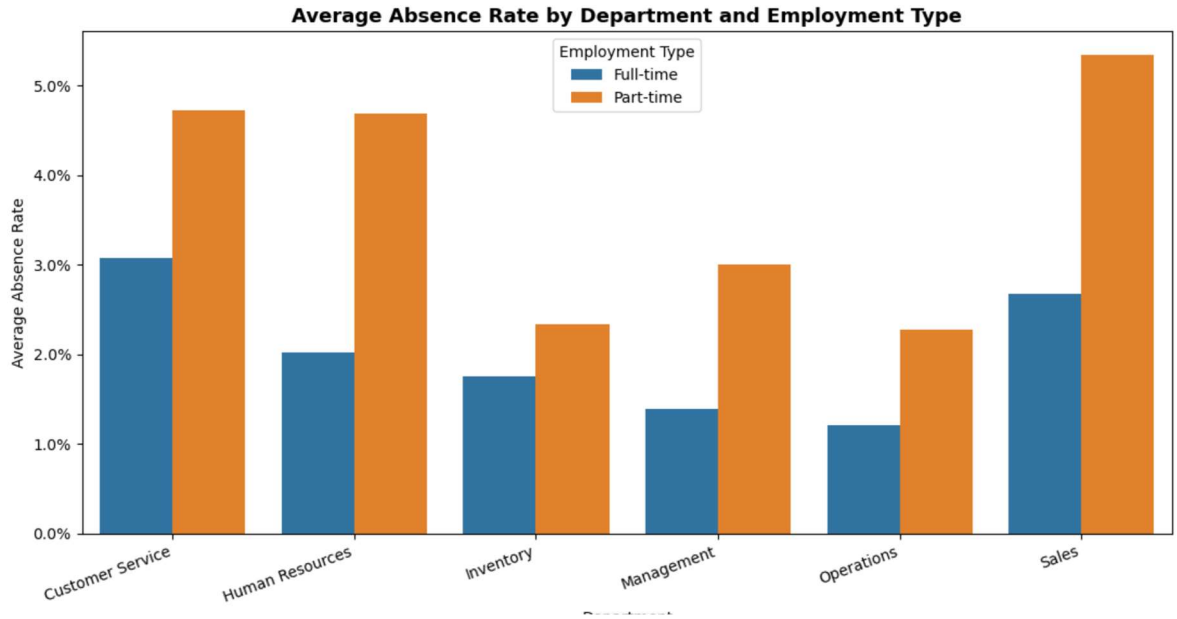


4.2 Department and Employment Type Comparison

Part-time employees had higher average absence rates in all six departments. Sales recorded the highest rates, particularly among part-time employees (approximately 5.3%). Human Resources and Customer Service also showed relatively high part-time absence rates.

These are descriptive differences and do not establish causation.

Figure 4.2 Average Absence Rate by Department and EmploymentType.



5. Correlation Analysis

5.1 Pearson and Spearman Correlations

Variable	Pearson r	Spearman r
AgeYears	-0.065	-0.06
LengthOfServiceYears	-0.096	-0.1
ScheduledHours2025	-0.232	-0.17
AbsentHours	0.934	0.98

Age and length of service showed very weak negative associations with absence rate. Scheduled hours also showed a weak negative association. Absent hours had a very strong positive relationship with absence rate.

The strong relationship between AbsentHours and AbsenceRate is partly mathematical because absence rate is calculated using absence hours. Pearson measures linear relationships, while Spearman measures monotonic relationships. The broadly consistent results suggest that the demographic relationships are not strongly dependent on the correlation method.

Figure 5.1.1 Employee Absenteeism – Correlation Heatmap

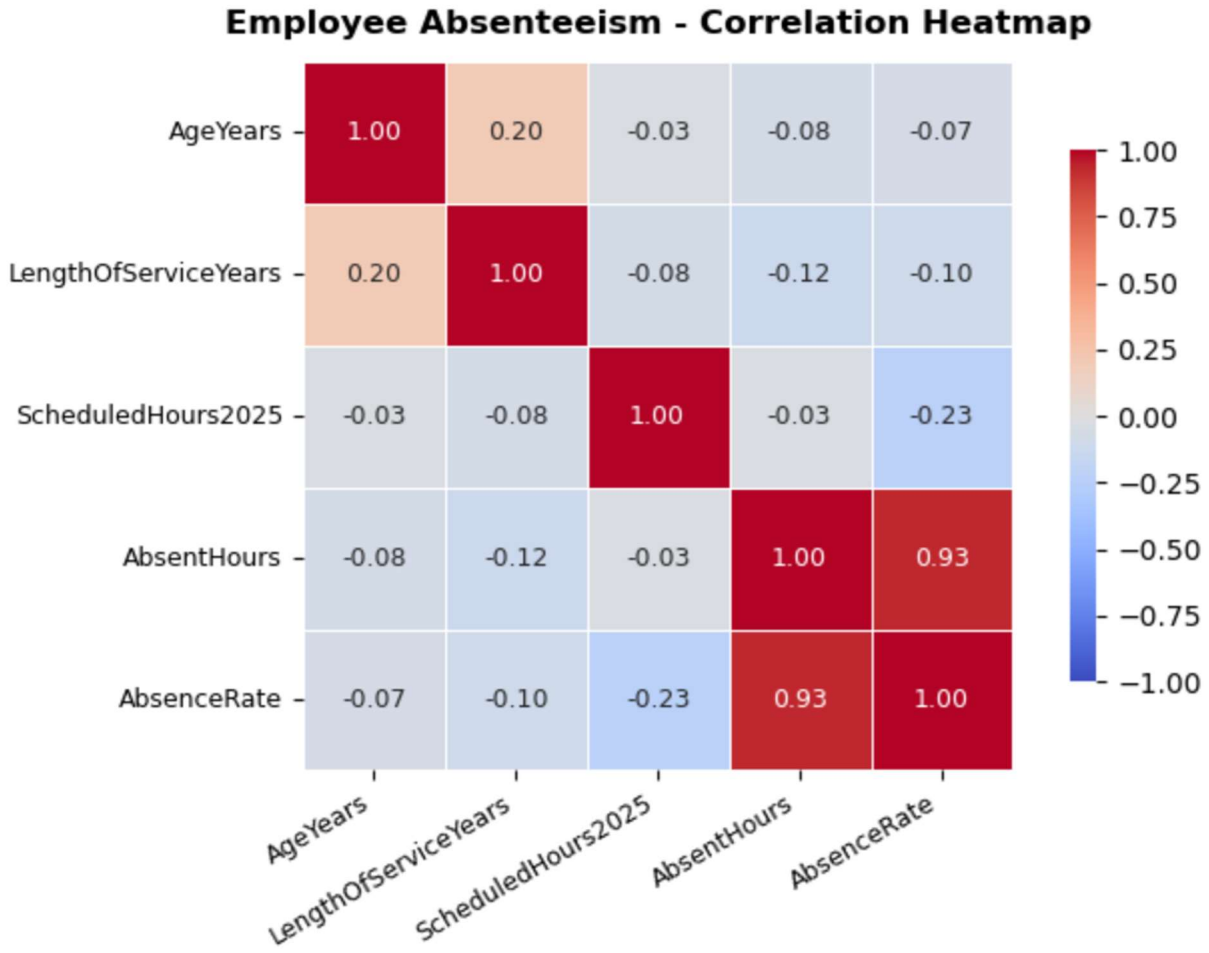
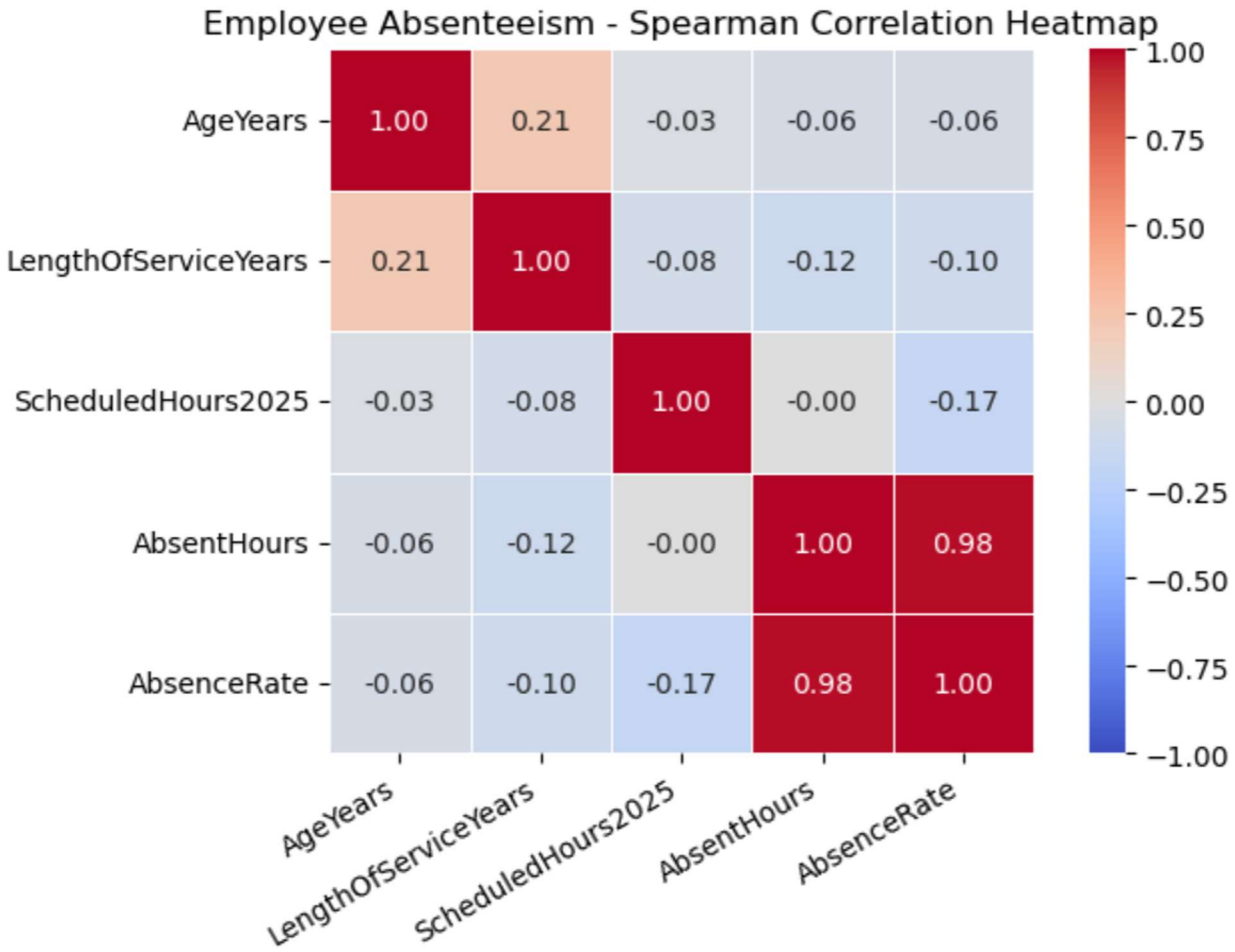


Figure 5.1.2: Spearman Correlation Heatmap of Employee Variables



5.2 Scatter Plot and Significance Analysis

The age scatter plot showed a very weak negative relationship with absence rate (r = -0.07). The length-of-service scatter plot showed a similar weak negative relationship (r = -0.10). Most employees had absence rates below 10%, although several exceeded 30%.

Relationship	Pearson r	Spearman r	Result
Age and absence rate	-0.0653	-0.0576	Both p < 0.05
Length of service and absence rate	-0.096	-0.1046	Both p < 0.05

Both relationships were statistically significant at the 5% level. However, the correlation coefficients were close to zero, indicating very weak associations. The age Spearman p-value (0.046360) was particularly close to the significance threshold. Multiple-testing considerations should therefore be taken into account.

Figure 5.2.1: Relationship Between Employee Age and Absence Rate

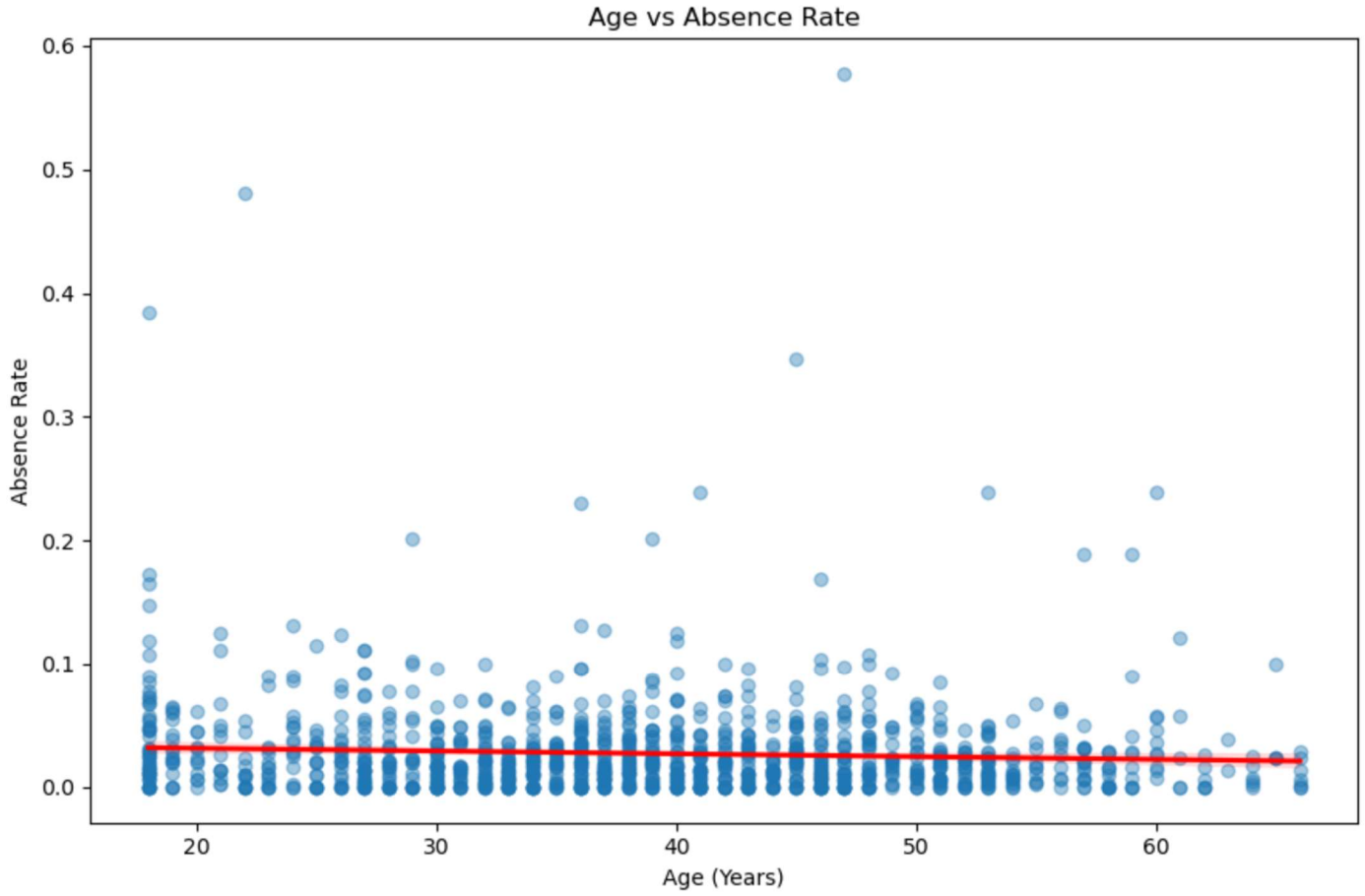


Figure 5.2.2: Relationship Between Length of Service and Absence Rate

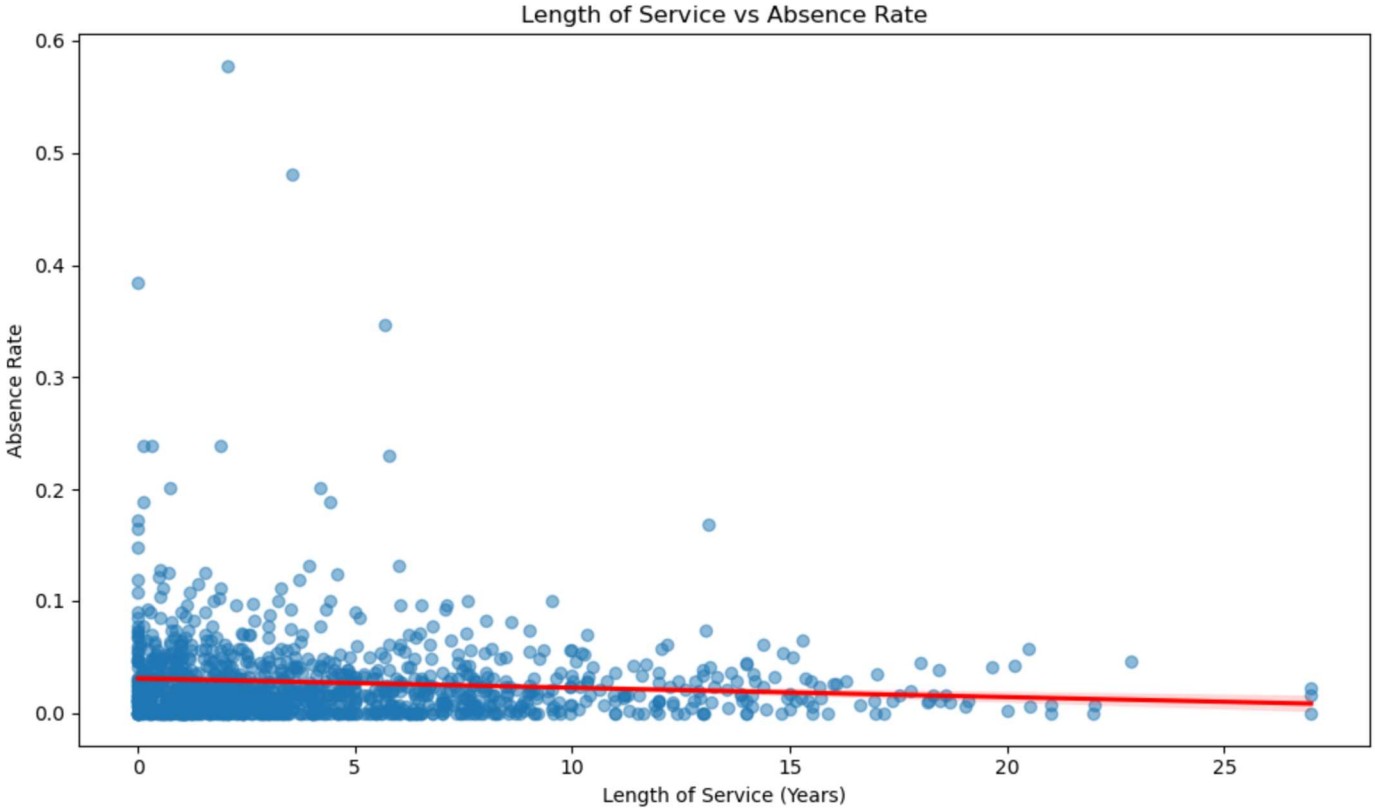


Figure 5.2.3 Correlation Significance Test

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CORRELATION SIGNIFICANCE TEST
-----------------------------------------

AgeYears vs AbsenceRate
Pearson correlation: -0.0653
Pearson p-value: 0.023795
Spearman correlation: -0.0576
Spearman p-value: 0.046360
Pearson result: Statistically significant

LengthOfServiceYears vs AbsenceRate
Pearson correlation: -0.0960
Pearson p-value: 0.000880
Spearman correlation: -0.1046
Spearman p-value: 0.000287
Pearson result: Statistically significant
```

Conclusion: Age and length of service have limited standalone predictive value for absenteeism, despite their statistical significance.

6. Outlier Detection and High-Absence Analysis

6.1 Absence Rate IQR Analysis

Measure	Result
Upper boundary	7.91%
Lower boundary	-3.82%
Median absence rate	1.73%
High-absence outliers	63
Percentage of valid employees	5.26%

A total of 63 employees exceeded the upper boundary. There were no lower outliers because absence rates cannot be negative. These are statistical outliers, not necessarily problematic employees.

Figure 6.1.1

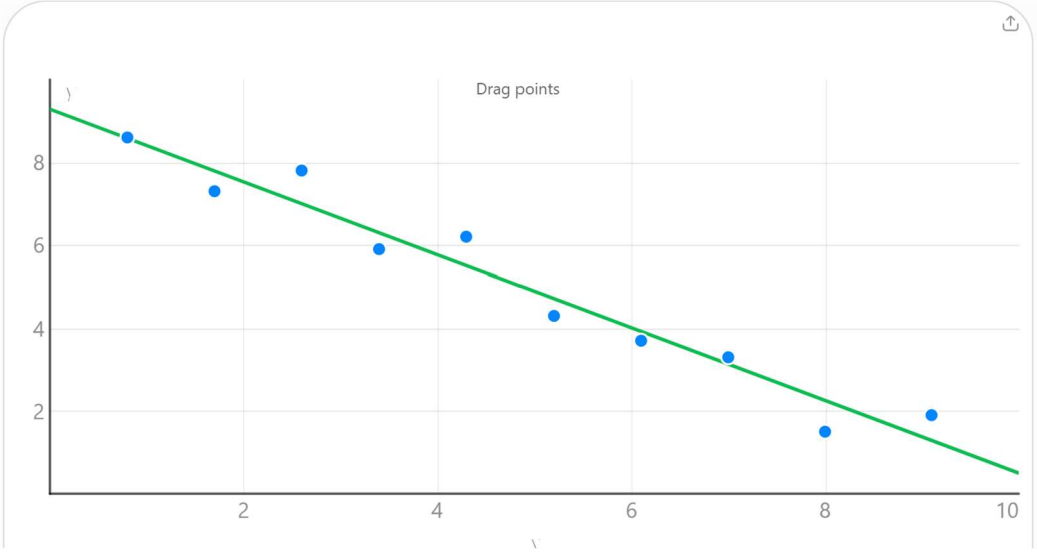


Figure 6.1.2

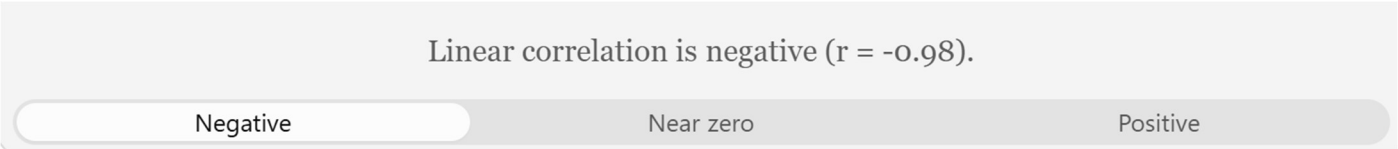


Figure 6.1.3 Absence Rate Outlier Detection

```
ABSENCE RATE OUTLIER DETECTION
------------------------------------
Q1: 0.0058 (0.58%)
Q3: 0.0351 (3.51%)
IQR: 0.0293
Lower Bound: -0.0382
Upper Bound: 0.0791

HIGH-ABSENCE OUTLIERS
------------------------------------
Number of potential outliers: 63
Percentage of employees: 5.26%
```

6.2 Departmental Distribution

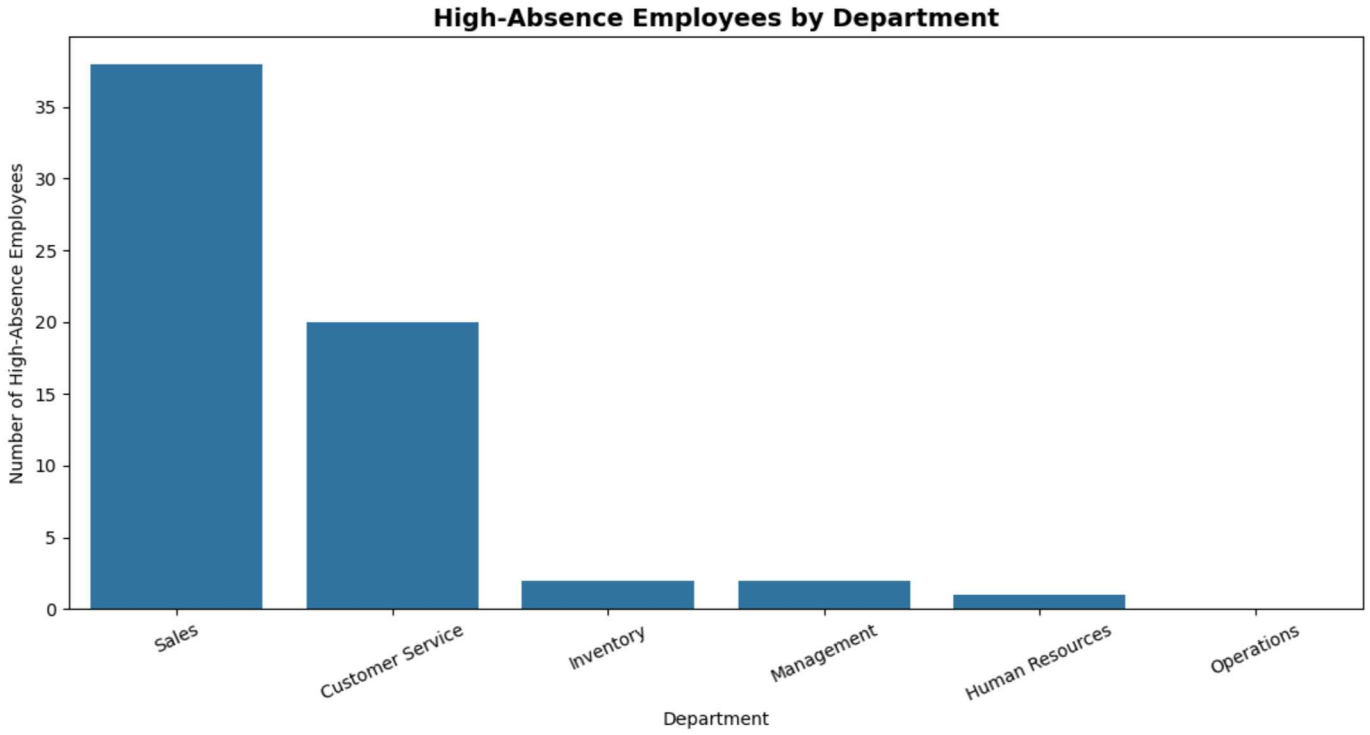
Department	Key finding
Sales	38 high-absence employees
Customer Service	Highest outlier rate: 10.47%
Inventory, Management and HR	Outlier rates: 0.88%–1.55%
Operations	Zero outliers among 139 employees

Sales accounted for 60.3% of all 63 high-absence cases, while Customer Service had the highest relative outlier rate. This highlights the distinction between the absolute number of cases and the proportion of employees affected.

Figure 6.3.1 High-Absence Employees by Department

```
HIGH-ABSENCE EMPLOYEES BY DEPARTMENT
------------------------------------------------------------------
                 Total_Employees  High_Absence_Count  Outlier_Percentage
Department
Sales                        443                  38                8.58
Customer Service             191                  20               10.47
Inventory                    228                   2                0.88
Management                   129                   2                1.55
Human Resources               68                   1                1.47
Operations                   139                   0                0.00
```

Figure 6.3.2 High-Absence Employees by Department



6.3 Employment Type Comparison

Employment type	High-absence cases	Total employees	Outlier rate
Full-time	24	862	2.78%
Part-time	39	336	11.61%
Total	63	1,198	5.26%

The three priority workforce segments accounted for 14 + 21 + 17 = 52 cases, representing 52 / 63 = 82.54% of all high-absence cases.
Customer Service part-time employees had the highest outlier rate, at 25.00%. Operations had no identified high-absence outliers among either employment type.

Figure 6.3.1 High- Absence Analysis by Department ad Employment Type

HIGH-ABSENCE ANALYSIS BY DEPARTMENT AND EMPLOYMENT TYPE

Department	EmploymentType	Total_Employees	High_Absence_Count	Outlier_Percentage
Customer Service	Full-time	135	6	4.44
	Part-time	56	14	25.00
Human Resources	Full-time	50	0	0.00
	Part-time	18	1	5.56
Inventory	Full-time	154	1	0.65
	Part-time	74	1	1.35
Management	Full-time	97	0	0.00
	Part-time	32	2	6.25
Operations	Full-time	97	0	0.00
	Part-time	42	0	0.00
Sales	Full-time	329	17	5.17
	Part-time	114	21	18.42

Figure 6.3.2 High- Absence Outlier Rate by Department and Employment Type

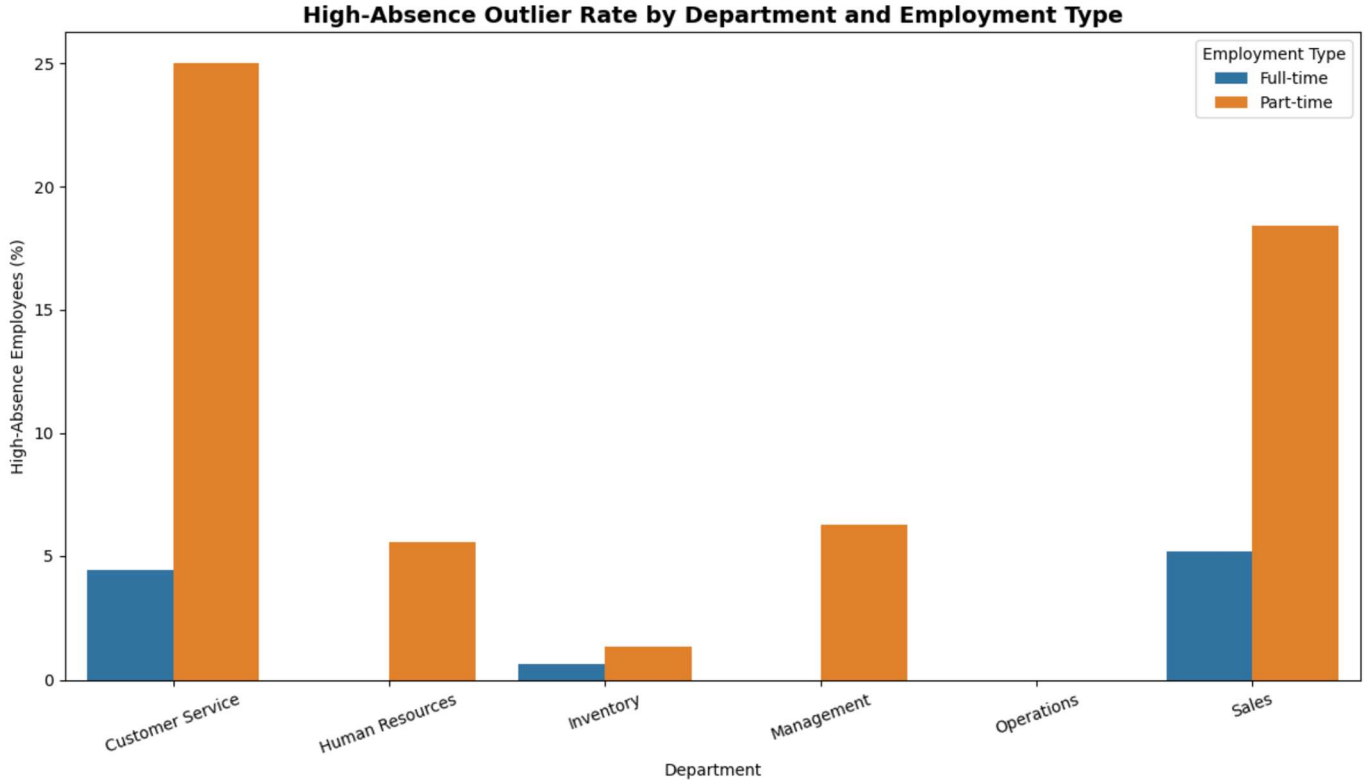


Figure 6.3.3 Contingency Table

```
CONTINGENCY TABLE
--------------------------------------------------
HighAbsence     False  True
EmploymentType
Full-time         838     24
Part-time         297     39

CHI-SQUARE TEST RESULTS
--------------------------------------------------
Chi-square statistic: 37.7739
P-value: 0.0000000008
Degrees of freedom: 1
Cramer's V: 0.1776

OUTLIER RATES BY EMPLOYMENT TYPE
EmploymentType
Full-time     2.78
Part-time    11.61
dtype: float64

Conclusion: Statistically significant association.
```

7. Statistical Significance Testing

7.1 Welch's Independent-Samples t-Test

Welch's t-test examined whether mean absence rates differed between full-time and part-time employees.

Statistical measure	Result
t-statistic	-5.2437
Degrees of freedom	396.66
p-value	< 0.001
Mean difference	1.71 percentage points
95% confidence interval	1.07–2.35 percentage points
Cohen's d	0.4472
Hedges' g	0.4469

Part-time employees recorded a mean absence rate of 3.94% (SD = 5.73%), compared with 2.23% (SD = 2.75%) for full-time employees. The difference was statistically significant, with a small-to-medium effect size.

The null hypothesis was rejected. Employment type was associated with differences in average absence rates, although causation cannot be established.

Figure 7.1.1: t-Test and Effect Size Analysis

Welch's t-test Results

Metric	Result
Full-time Sample Size	862
Part-time Sample Size	336
T-statistic	-5.2437
P-value (displayed value)	0.0

Analysis Summary

Analysis Metric	Result
Full-time Average Absenteeism Rate	2.23%
Part-time Average Absenteeism Rate	3.94%
Mean Difference	1.71 percentage points
Welch's t-statistic	-5.2437
P-value	p < 0.001
Cohen's d	0.4472

Figure 7.1.2

Statistical Summary

Statistical Metric	Result
Full-time Average Absenteeism Rate	2.23%
Part-time Average Absenteeism Rate	3.94%
Mean Difference	1.71 percentage points
95% Confidence Interval	1.07–2.35 percentage points
Welch's t-statistic	-5.2437
Welch's Degrees of Freedom	396.66
P-value	p < 0.001
Cohen's d	0.4472
Hedges' g	0.4469

7.2 Chi-Square Test: Employment Type and High-Absence Status

Hypotheses:

- H_0: Employment type and high-absence status are independent.
- H_1: Employment type and high-absence status are associated.

The Chi-square test identified a statistically significant association, with a p-value substantially below 0.05. The null hypothesis was rejected.

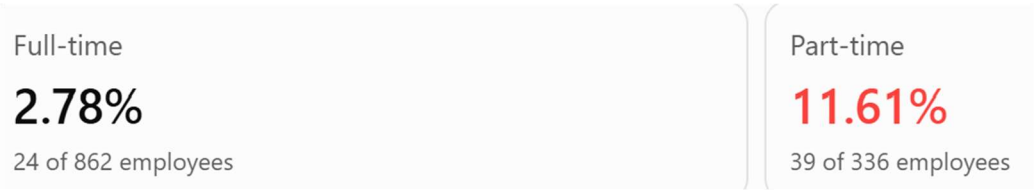
The part-time outlier rate (11.61%) was approximately 4.18 times the full-time rate (2.78%). However, Cramér's V was 0.1776, indicating a relatively modest association in effect-size terms.

Figure 7.2.1 Chi-Square Test Results

Chi-Square Test Results

Statistical Measure	Result
Chi-square Statistic (χ^2)	37.7739
Degrees of Freedom (df)	1
P-value	0.0000000008
Cramér's V	0.1776
Significance Level (α)	0.05
Statistical Conclusion	Significant Association

Figure 7.2.2 Business significance



7.3 Statistical Considerations

The overall Chi-square test does not control for departmental differences. Since Customer Service and Sales have particularly high part-time outlier rates, department composition could partly explain the overall association.

Further stratified analysis or multivariable modelling is recommended before concluding that employment type independently influences absenteeism.

8. Management Recommendations and Conclusion

8.1 Management Recommendations

1. Departmental investigation: Prioritise Customer Service and Sales because of their relatively high absence rates and outlier concentrations.
2. Part-time workforce review: Examine scheduling, workload and working arrangements, particularly in Customer Service and Sales.
3. Individual case assessment: Review the 63 employees exceeding the 7.91% IQR threshold, considering legitimate absence reasons and employee circumstances.

4. New employee support: Strengthen onboarding, induction and early-stage engagement for employees with less than two years of service.
5. Data quality improvement: Standardise department names, monitor missing attendance records and maintain consistent absence calculations.
6. Further statistical analysis: Use stratified analysis or multivariable regression to examine the independent contributions of employment type, department, age and length of service.

8.2 Overall Conclusion

The 2025 EDA demonstrates that absenteeism is not evenly distributed across Mega Supermart.
Customer Service and Sales recorded the highest departmental weighted absence rates. Part-time employees had higher average absence rates and were disproportionately represented among high-absence outliers.
Most notably, 82.54% of identified high-absence cases were concentrated in Customer Service part-time, Sales part-time and Sales full-time employees.
Statistical testing confirmed significant differences in average absence rates between employment types and an association between employment type and high-absence status. However, departmental differences and other workforce characteristics must be considered.
Age and length of service showed statistically significant but weak negative associations with absenteeism.

Final management perspective: Mega Supermart should adopt a targeted, evidence-based approach to attendance management, focusing on workforce segments with higher absence concentrations, improving employee support and strengthening HR data governance.
Statistical outliers should trigger further investigation, not automatic assumptions of misconduct.

